

PRELIMINARY SPECIFICATION DRAFT

Challenge: During the integration phase, there was a hardware bottleneck. The three directional sensors and underside sensor and the motor drivers exhausted all available digital IO pins on the arduino nano.

Solution: I researched the ATmega328P silicon architecture and confirmed that pins A0 through A5 are internally mapped and can be addressed as digital pins using the “pinMode()” and “digitalRead()” functions

I instructed the students to solder 12 header pins (arranged in 4 signal-power-ground groups) onto the prototype shield, specifically utilizing the analog rail for digital input and pins for easy replacement incase of sensor failure

I guided the students through updating their code logic to treat A1-A5 as digital triggers successfully expanding our sensor capacity without needing a larger microcontroller

